

CS4048 – Data Science (BCS-7B, BCS-7C)

Assignment#1

Due Date: September 10, 2024

Instructions:

- Submit your assignment through Google Classroom by the deadline.
 - Submit your code in the form of a **Jupyter/Google-Colab notebook** with the filename format **roll_number.ipynb**, along with a **PDF report** explaining your approach and findings.
 - Ensure your code is efficient and written in a Pythonic style—avoid long, verbose code typical of C++.
 - **Plagiarism** is strictly prohibited. Any instance of plagiarism will result in zero marks for all assignments.
 - You must understand your code and its functionalities thoroughly, as a **viva** will be conducted during evaluation.
 - Focus on demonstrating your understanding and skills rather than copying from others.
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Task 1: Static Website Extract Information About the CodeTechMentor Creator

URL: <https://www.codetechmentor.com/p/about-me.html>

Write a Python script that uses the **requests** library to fetch the HTML content of the **CodeTechMentor** about page. Use **BeautifulSoup** to parse the HTML and extract the following details about the creator:

1. **Creator's Name:** Extract the name of the **CodeTechMentor** creator.
2. **Introduction Text:** Retrieve and combine all the introductory text.
3. **Social Media Links:** Find social media links.

Print the results in a well-structured format to easily view the output. Ensure your script handles cases where elements might be missing and provides appropriate messages if any of the required information is not found.

```
Mannan Ul Haq
Associate Data Scientist
I'm a Data Science student at FAST-NUCES, Lahore, deeply passionate about data science, AI, and machine learning. I thrive on diving into data, uncovering patterns, and tackling complex problems.
https://www.facebook.com/mannan.ulhaq.5/
https://www.instagram.com/urs__maan/
https://www.linkedin.com/in/mannan-ul-haq-5578b721b/
https://github.com/MannanUlHaq
```

Task 2: Static Website Data Extraction (PriceOye Apple Laptops)

URL: <https://priceoye.pk/laptops/apple>

Write a Python script that uses the **requests** library to retrieve the HTML content of the **PriceOye** Apple laptops webpage. Use **BeautifulSoup** to parse the HTML and extract the **names**, **prices**, and **discounts** of all Apple laptops listed. Ensure to exclude any products marked as **discontinued**. Print the results in a well-structured format to easily view the output.

```
Apple Macbook Pro 14 MTL73 (8GB-512GB)
Rs 452,999
9% OFF

Apple MacBook Air 13 M1 MGN63 (8GB-256GB)
Rs 242,999
6% OFF

Apple Macbook Pro MRX33 M3 Pro 14.2 (18GB-512GB)
Rs 549,999
8% OFF

Apple MacBook Pro 14.2" MTL83 (8GB-1TB)
Rs 494,999
4% OFF

Apple MACBOOK AIR 13.6" M2 CHIP MLY43 8-512GB
Rs 354,999
11% OFF

Apple Macbook Pro MRX33 M3 Pro 14 (18GB-512GB) Black
Rs 554,999
8% OFF

Apple MACBOOK AIR 13" M3 CHIP MRXV3 8-256
Rs 334,999
16% OFF

Apple MacBook Air 15.3 M2 MQKX3 (8GB-512GB)
Rs 406,999
No discount found
```

Task 3: Dynamic Website Data Extraction (Zameen.com Property Listings)

URL: https://www.zameen.com/Houses_Property/Lahore_DHA_Defence-9-1.html

Using **Selenium**, navigate through the first **five** pages of property listings on **Zameen.com** for houses in DHA, Lahore. Extract details such as **location**, **price**, **number of beds**, **baths**, and **area**. Parse the HTML content using **BeautifulSoup** and format the data in a **pandas** DataFrame.

		Location	Price	Beds	Baths	Area
0		DHA Phase 7, DHA Defence	7.85 Crore	5	6	1 Kanal
1		DHA Phase 7, DHA Defence	6.7 Crore	5	6	1 Kanal
2	DHA Phase 6 - Block J,	DHA Phase 6	7.9 Crore	5	6	1 Kanal
3	DHA Phase 6 - Block D,	DHA Phase 6	8.55 Crore	5	6	1 Kanal
4		DHA 9 Town, DHA Defence	3.65 Crore	4	5	8 Marla
..		...	...	...	...	...
130		DHA Phase 6, DHA Defence	7.65 Crore	5	6	1 Kanal
131	DHA Phase 6 - Block K,	DHA Phase 6	9.75 Crore	5	6	1 Kanal
132		DHA Phase 6, DHA Defence	10.75 Crore	5	6	1 Kanal
133	DHA Phase 6 - Block L,	DHA Phase 6	13.5 Crore	5	6	1 Kanal
134	DHA Phase 5 - Block J,	DHA Phase 5	7.99 Crore	5	6	1 Kanal
[135 rows x 5 columns]						

Task 4: Dynamic Website Data Extraction (ESPNcricinfo Cricketer Stats)

In this task, you'll use **Selenium** to automate web browsing and scrape dynamic data from a website. The website we will extract data from is **ESPNcricinfo**, specifically a page with cricketer statistics. The steps for this task are as follows:

1. Navigate to Google:

- First, open Google (<https://www.google.com/>) using Selenium.
- You'll use the **Selenium** to locate the search box.
- In the search box, enter the phrase: "**Best economy rates for Pakistan in Tests**" using Selenium **send_keys** method.
- Simulate pressing the Enter key using **Keys.ENTER**.

2. Click on the first search result:

- Once the search results appear, use **Selenium** to identify and click on the first search result. The first result should direct you to the **ESPNcricinfo** page with bowling stats for Pakistan.
- The website link is: <https://www.espnecricinfo.com/records/team/bowling-best-career-economy-rate/pakistan-7/test-matches-1>

3. Extract cricketer statistics:

- After reaching the ESPNcricinfo page, use **Selenium** to locate the table that contains the cricketer statistics.
- The table should include columns like **player names, matches played, runs given, wickets taken, and economy rates**.

4. Parse and format the data:

- Extract the data from the table.
- After extracting the data, store it in a **pandas DataFrame**.
- Ensure that each row corresponds to a player and each column to a specific statistic.

5. Output the DataFrame:

- Display the resulting DataFrame

	Player	Span	Mat	Overs	Mdms	Balls	Runs	Wkts	BBI	Ave	Econ	SR	4	5	10
0	Pervez Sajjad	1964-1973	19	-	217	4145	1410	59	7/74	23.89	2.04	70.2	4	3	-
1	Mohammad Nazir	1969-1983	14	543.4	162	3262	1124	34	7/99	33.05	2.06	95.9	-	3	-
2	Shujauddin	1954-1962	19	385.3	121	2313	801	20	3/18	40.05	2.07	115.6	-	-	-
3	Fazal Mahmood	1952-1962	34	1639.0	560	9834	3434	139	7/42	24.70	2.09	70.7	6	13	4
4	AH Kardar	1952-1958	23	452.0	139	2712	954	21	3/35	45.42	2.11	129.1	-	-	-
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
111	Shahid Nazir	1996-2007	15	372.2	71	2234	1272	36	5/53	35.33	3.41	62.0	-	1	-
112	Wahab Riaz	2010-2018	27	836.2	118	5018	2864	83	5/63	34.50	3.42	60.4	3	2	-
113	Umar Gul	2003-2013	47	1599.5	256	9599	5553	163	6/135	34.06	3.47	58.8	12	4	-
114	Mohammad Sami	2001-2012	36	1249.5	193	7499	4483	85	5/36	52.74	3.58	88.2	3	2	-
115	Naseem Shah	2019-2024	18	506.3	84	3039	1825	54	5/31	33.79	3.60	56.2	2	1	-

116 rows × 15 columns

Task 5: Dynamic Website Data Extraction (YouTube Video Content - Cristiano Ronaldo)

In this task, you will use **Selenium** and **BeautifulSoup** to extract video information from Cristiano Ronaldo's official YouTube channel. Specifically, you will gather the **video titles**, **view counts**, and **comments counts**. The steps involved are as follows:

- 1. Navigate to the YouTube Channel:**
 - Use Selenium to open the YouTube URL:
<https://www.youtube.com/@cristiano/videos>
 - Ensure that the page fully loads before proceeding to extract data.
- 2. Extract Video Titles and View Counts:**
 - Use **BeautifulSoup** to parse the page content.
 - Locate the section containing the list of videos. Extract the titles of each video and their corresponding view counts.
- 3. Navigate to Each Video Page for Comments Count:**
 - For each video, use Selenium to click on the video link, which will open the video in a new tab.
 - Since the comments section loads dynamically as you scroll, use Selenium's **scrolling method** to scroll down the page until the comments section is visible.
- 4. Extract the Comments Count:**
 - Once the comments section is loaded, use **BeautifulSoup** or **Selenium** to locate and extract the number of comments on the video.
- 5. Print the Results:**
 - After collecting all the data (video titles, views, comments count), print the results in a well-structured format.

```
Title: My thoughts on my 2023/24 season and the truth about my future. Talk with my friend Rio (PART 1)
Views: 3.2M views
Comments: 60,087

Title: Georgina RATES MY OUTFITS from the last 20 years...You won't believe which one is her favorite!
Views: 3.2M views
Comments: 111,195

Title: "Bernabéu... pressure... 90,000" 🔥🔥🔥
Views: 3.2M views
Comments: 173,799

Title: The golden button... for my golden kids
Views: 3.2M views
Comments: 239,915

Title: This is how I felt when I discovered paradise... 🌊 THE RED SEA Project, Saudi
Views: 3.2M views
Comments: 64,891
```